Shock Engine: Conditional Post-Shock Continuation in Bitcoin and Ether

Design, falsification and residual alpha of a systematic intraday strategy

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Abstract

We study a systematic strategy that trades the continuation of abnormal 15-minute price moves in Bitcoin (BTC) and Ether (ETH). A position is opened when a return that is extreme relative to its local volatility breaks the recent range and is confirmed by higher-timeframe structure, candle quality, participation and a price-impact filter; risk is managed with volatility-scaled exits. New positions are opened only when the daily volatility regime is calm, and the short side is a *conditional mirror* of the long side: short entries reproduce the long logic in the opposite direction and are admitted only when the daily trend regime is bearish. The parameters were calibrated on Bitcoin and applied unchanged to Ether; the two entry conditions were set with knowledge of both markets. Over 1 September 2018 to 30 September 2026, a 50/50 BTC/ETH portfolio shows a Sharpe ratio of 2.02, a compound annual growth rate of 67.7% and a maximum drawdown of −17.8% after transaction costs, against 0.78, 34.1% and −76.9% for buy-and-hold. Random-entry placebos, delayed entries, parameter neighbourhoods, cross-asset transfer, a Monte Carlo falsification battery for the short-entry condition and a regression on six standard trend-following strategies (residual alpha of 44.8% a year, Newey–West $t = 4.94$) do not explain the result away. The evidence is nonetheless in-sample: the parameters were set on Bitcoin data covering the period and the entry conditions with knowledge of both markets. We report the selection risk of the rule choice (probability of backtest overfitting 19%), the failures (TAO; partial results on Solana and gold; a walk-forward Sharpe of 0.83) and pre-register a forward test.

Keywords: cryptocurrency, intraday momentum, time-series momentum, jumps, regime conditioning, backtest overfitting, pre-registration, residual alpha.
JEL: G11, G12, G14, C12, C58.

Important. All performance figures in this paper come from a historical simulation of fixed rules after modelled transaction costs. They are not live trading results. Simulated results have inherent limitations: they benefit from hindsight in the design of the rules and do not reflect the liquidity, slippage, funding and market impact of a real account. Past simulated performance does not predict future results. Nothing in this document is investment advice or an offer of any financial product or service.

1 Introduction

Cryptocurrency markets trade continuously, are dominated by a small number of liquid perpetual and spot venues, and move in bursts. A large share of daily variance arrives in a few intraday intervals, often after news, liquidations or the release of resting liquidity. Two competing views predict what happens next. Under *underreaction*, information is incorporated gradually and a large move is followed by drift in the same direction (Hong and Stein, 1999). Under *overreaction* or liquidity provision, the move is partly a temporary price concession that reverts. Which view dominates is an empirical question, and the answer may depend on the state of the market.

This paper documents a rule-based strategy, *Shock Engine*, built on the first view: it enters in the direction of an abnormal 15-minute move once the move has been qualified by a small set of filters, and exits through volatility-scaled risk management. Our contribution is less the rule itself than the way it is tested. We make five points.

(i) The parameters were set on Bitcoin and transferred *without recalibration* to Ether, where they meet every pre-set criterion on two independent price feeds; on other markets they are partial on Solana and gold (strategy criteria met, event study not) and fail on TAO, and we report these results.

(ii) Random-entry placebos with identical exits and one-bar entry delays show that the edge lies in the timing of entries rather than in market drift or in the shape of the exits.

(iii) New positions are opened only in the calm volatility regime, and the short side is a *conditional mirror* of the long side: short entries are allowed only in a bearish daily trend regime. Both conditions were chosen on the same data that are used to evaluate them, so we measure the probability that the choice among the rule variants we tried is overfitted, and subject the short condition to three Monte Carlo falsification tests specified before they were run.

(iv) A regression of the daily returns on buy-and-hold and on six standard trend-following strategies leaves 81% of the average return unexplained, with a Newey–West t-statistic of 4.94.

(v) We pre-register a forward test of the short-entry condition with interim analyses, and run the engine on Hyperliquid's test network so that its decisions are timestamped by a third party.

The headline numbers are strong, which is precisely why they deserve suspicion. Throughout, we try to separate what the data can establish (the result is not a simple artefact of drift, exits, costs, a single parameter choice or a generic trend factor) from what it cannot (that the result will persist out of sample).

2 Related literature

Momentum and trend following. Cross-sectional momentum in equities was documented by Jegadeesh and Titman (1993); Moskowitz et al. (2012) show that an asset's own past return predicts its future return across futures markets (time-series momentum), and Hurst et al. (2017) extend the evidence over a century. Momentum payoffs are state dependent and can crash in rebounds after market declines (Daniel and Moskowitz, 2016), which motivates conditioning the short side on the state of the market rather than treating it as a symmetric copy of the long side. Simple trend rules based on moving averages and range breakouts were evaluated against bootstrap null models by Brock et al. (1992); moving-average timing filters are popularised by Faber (2007).

Intraday dynamics and jumps. Gao et al. (2018) show that the first half-hour return of the S&P 500 ETF predicts its last half-hour return, a form of intraday momentum linked to late-informed and hedging flows. Our shock statistic standardises a short-horizon return by an estimate of local volatility, in the spirit of the nonparametric jump test of Lee and Mykland (2008). Jumps cluster in time: self-exciting point processes (Hawkes, 1971) capture this clustering and its propagation across markets (Aït-Sahalia et al., 2015), and one of our filters measures the local intensity of shocks. Scaling exposure and exits with volatility relates to the volatility-managed portfolios of Moreira and Muir (2017).

Cryptocurrency returns. Liu and Tsyvinski (2021) find that cryptocurrency returns load on crypto-specific factors, notably momentum and investor attention, rather than on traditional asset factors; Liu et al. (2022) propose a three-factor model (market, size, momentum) for the cross-section of cryptocurrencies. Our strategy operates at a much shorter horizon, and Section 10 measures how much of it a set of standard trend factors can span.

Overfitting and inference. Searching many rules on one history inflates in-sample performance. White (2000) and Hansen (2005) propose reality checks for data snooping; Bailey and López de Prado (2014) deflate the Sharpe ratio for the number of trials and non-normality; Bailey et al. (2017) estimate the probability of backtest overfitting; Harvey et al. (2016) argue for a t-statistic hurdle of about 3 for new factors. We use heteroskedasticity- and autocorrelation-consistent standard errors (Newey and West, 1987), block bootstraps (Künsch, 1989; Politis and Romano, 1994), the Sharpe-ratio standard error of Lo (2002), and pre-registration with interim analyses in the tradition of clinical trials (Haybittle, 1971; Peto et al., 1976; Nosek et al., 2018).

3 Data and conventions

Bitcoin is BTC/USD on Bitstamp and Ether is ETH/USDT on Binance spot, both in 15-minute bars timestamped at the open in UTC. After a warm-up period, Bitcoin is simulated from January 2017 and Ether from September 2018. An independent Ether feed (ETH/USD, Dukascopy) is used for replication. The common period of the portfolio is 1 September 2018 to 30 September 2026 (2,952 days, 8.1 years).

Daily returns are close-to-close on UTC days, the close of a day being the close of its last 15-minute bar. Annualisation uses $\sqrt{365.25}$ and a zero risk-free rate. Every order pays a commission of 0.045% of notional, the base taker fee of the intended execution venue. Slippage and perpetual funding are *not* included in the headline figures; Section 6.4 adds them. Each market trades 100% of its own capital, one position at a time, without leverage.

4 The strategy

We describe the logic of the rules and the mathematics of their building blocks. Thresholds, window lengths, timeframes and exit multiples are proprietary and are not disclosed; none of the tests below depends on knowing them. The rules described here are those reported on the public website; the pre-registration of Section 11 covers the short-entry condition.

4.1 Shock detection

Let P_t be the close of 15-minute bar t, O_t, H_t, L_t its open, high and low, and $r_t = \ln(P_t/P_{t-1})$. Rolling estimates of the mean $\hat{\mu}_t$ and standard deviation $\hat{\sigma}_t$ of r over a trailing window give the

standardised return

$$z_t = \frac{r_t - \hat{\mu}_t}{\hat{\sigma}_t}, \qquad S_t = \mathbf{1}\{|z_t| > \kappa\}. \tag{1}$$

An up-shock ($S_t = 1$, $r_t > 0$) is directional when the close also breaks the range of the preceding bars, $P_t > \max_{t-h \le s < t} H_s$; a down-shock is the mirror image. The standardisation makes the trigger adaptive: the same absolute move is a shock in a calm market and noise in an agitated one, as in local-volatility jump tests (Lee and Mykland, 2008).

4.2 Qualification

A directional shock becomes an entry only if the following filters agree.

(a) *Structure.* Price is on the side of a higher-timeframe moving average that corresponds to the trade; long entries also require the average to be rising.

(b) *Candle quality.* The shock bar closes near its extreme with a small counter-direction wick, e.g. for a long $(H_t - \max(O_t, P_t))/(H_t - L_t) < \omega$, so that the move was not rejected within the bar.

(c) *Participation.* Volume is abnormal relative to its own recent distribution, $(V_t - \bar{V}_t)/s_{V,t} > \nu$.

(d) *Shock clustering.* The local intensity of shocks is estimated by an exponentially weighted average of past shock indicators,

$$\hat{\lambda}_t = (1 - \eta)\, \hat{\lambda}_{t-1} + \eta\, S_t, \tag{2}$$

and ranked against its own trailing distribution. Long entries require shocks to be clustering (a high percentile of $\hat{\lambda}_t$), the empirical signature of the self-exciting dynamics of jumps (Hawkes, 1971; Aït-Sahalia et al., 2015).

(e) *Volatility state.* The average true range, standardised against its own recent history, must lie below a regime-specific level, so that entries depend on the volatility state that precedes the shock.

A short cooling-off period follows each trade.

4.3 Volatility regimes

Daily realised volatility over a short window is ranked against its own trailing one-year distribution. The market is *calm* below the median and *agitated* above, and the regime is read from the last closed day, so that the rule never uses information from the current day. New positions are opened only when the market is calm; a position opened in a calm market keeps its exits if the regime turns agitated. The rules were first designed with a second parameter set for the agitated regime, which only allowed long entries; it no longer opens positions.

4.4 Exits and risk

At entry the strategy records the average true range A. The initial stop is placed at $P_{\text{entry}} \mp m_s A$. A position is otherwise closed when a qualified shock occurs in the opposite direction (flip exit); a stop that trails the best price by $m_r A$ is set wide and rarely triggers, and there is no partial take-profit. Each market holds at most one position, sized to its capital, with no pyramiding, no averaging down and no leverage.

4.5 The short side: a conditional mirror

Short entries reproduce the long logic in the opposite direction: a down-shock that breaks the recent low, closes near its low, occurs below the higher-timeframe average and is accompanied by abnormal volume. The long side carries additional qualification (rising structure and shock clustering), which reflects the asymmetry of crypto markets, whose long-run drift has been strongly positive.

The mirror is *conditional* on the daily trend regime. With $D(t)$ the last UTC day closed before bar t, P^{d}_D the daily close, $\bar{P}^{(n)}_D$ a medium-term moving average of daily closes and k a slope horizon in days,

$$T_t = \mathbf{1}\Big\{P^{\mathrm{d}}_{D(t)} < \bar{P}^{(n)}_{D(t)} \text{ and } \bar{P}^{(n)}_{D(t)} < \bar{P}^{(n)}_{D(t)-k}\Big\}, \qquad \mathrm{Short}_t = \mathrm{Mirror}_t \times T_t. \tag{3}$$

Long entries, exits and the flip rule do not depend on T_t; the condition only decides whether a new short position may be opened. The economic motivation is the state dependence of momentum (Daniel and Moskowitz, 2016): selling a downside shock is a continuation trade in a falling market and a counter-trend trade in a rising one. Over the sample, 250 of the 716 trades are shorts.

Design history. For most of the research the strategy traded both volatility regimes with an unconditional (symmetric) short side. The trend condition on shorts was adopted in October 2026 after a pre-registered study of short entries in which it was one of several candidate rules, not the primary hypothesis; the calm-regime restriction was adopted later the same month after an exploratory study of the agitated regime. Both were chosen with knowledge of the 2017–2026 Bitcoin and Ether data, so every historical figure in this paper is in-sample with respect to them. Section 8 measures the selection risk of the rule choice, Section 9 asks whether a trend condition chosen at random would have done as well, and Section 11 tests the short condition prospectively.

4.6 Portfolio

The portfolio allocates half of the capital to the Bitcoin strategy and half to the Ether strategy on the first day and never rebalances. Rebalancing monthly or weighting by risk changes the Sharpe ratio by at most 0.06 and is not part of the specification.

5 Statistical framework

Performance. With daily returns r_d, mean $\bar{r}$ and population standard deviation s,

$$\mathrm{SR} = \sqrt{365.25}\,\frac{\bar{r}}{s}, \qquad \mathrm{Sortino} = \sqrt{365.25}\,\frac{\bar{r}}{\sqrt{\frac{1}{N}\sum_d \min(r_d,0)^2}}, \qquad \mathrm{MDD} = \min_t\Big(\frac{V_t}{\max_{s\le t} V_s} - 1\Big), \tag{4}$$

and the Calmar ratio is CAGR/|MDD|. For independent daily returns, the standard error of a Sharpe ratio annualised from daily returns over T years is approximately (Lo, 2002)

$$\widehat{\mathrm{SE}}(\widehat{\mathrm{SR}}) \approx \sqrt{365.25}\,\sqrt{\frac{1+\frac{1}{2}\mathrm{SR}_d^2}{N}} \approx \frac{1}{\sqrt{T}}, \qquad \mathrm{SR}_d = \frac{\mathrm{SR}}{\sqrt{365.25}}, \quad N = 365.25\,T. \tag{5}$$

Risk-normalised trade returns. To compare trades across volatility regimes and assets, trade i with profit π_i, equity at entry E_i and entry day d_i is scored as

$$R_i = \frac{\pi_i/E_i}{\sigma_{d_i}}, \tag{6}$$

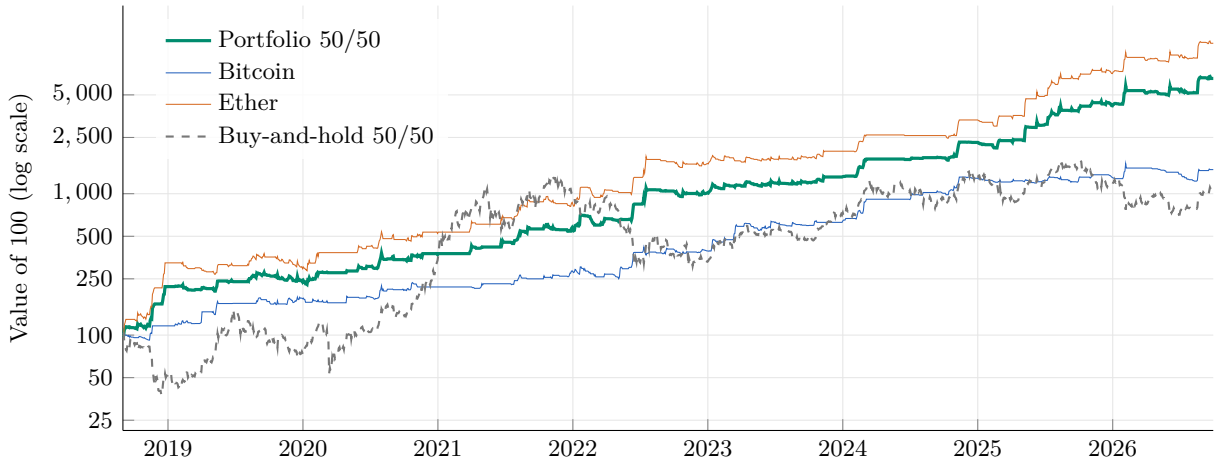


Figure 1: Equity curves, 1 September 2018 to 30 September 2026, daily closes. Historical simulation after modelled transaction costs; buy-and-hold without costs.

where σ_d is the standard deviation of the 30 daily log returns before day d. For a set of short trades, EV $= \overline{R}$, and the regime contrast is

$$\Delta = \overline{R}_{T=1} - \overline{R}_{T=0}. \tag{7}$$

Monte Carlo inference. For a statistic θ and N counterfactual draws θ^*_j, the one-sided Monte Carlo p-value is

$$p = \frac{1 + \#\{j : \theta^*_j \geq \theta\}}{1 + N}. \tag{8}$$

Residual alpha. Daily strategy returns are regressed on benchmark returns,

$$r_t = \alpha + \sum_{k=1}^{K} \beta_k f_{k,t} + \varepsilon_t, \tag{9}$$

by ordinary least squares, with the Newey–West covariance

$$\hat{V} = (X^\top X)^{-1}\Big[\hat{\Gamma}_0 + \sum_{l=1}^{L}\Big(1 - \frac{l}{L+1}\Big)(\hat{\Gamma}_l + \hat{\Gamma}_l^\top)\Big](X^\top X)^{-1}, \qquad \hat{\Gamma}_l = \sum_{t=l+1}^{T} \hat{\varepsilon}_t \hat{\varepsilon}_{t-l}\, x_t x_{t-l}^\top, \tag{10}$$

and lag $L = \lfloor 4(T/100)^{2/9} \rfloor$. The residual information ratio is $\sqrt{365.25}\,\hat{\alpha}/\hat{\sigma}_\varepsilon$. A moving-block bootstrap draws calendar months with replacement, the same months for all series, so that cross-sectional dependence is preserved (Künsch, 1989).

6 Results

6.1 Headline performance

Table 1 reports the two markets, the portfolio and a buy-and-hold benchmark built in the same way (half of the capital in each coin on the first day, no rebalancing, no costs). Figure 1 shows the equity curves on a logarithmic scale.

The return distribution is strongly right-skewed: the strategy is in the market 25% of the time, wins 31% of its trades, and its average winner is 5.6 times its average loser. The best 5% of trades account for 99% of the portfolio's log growth; without them the CAGR would be −0.3%. This is the profile of a convex, trend-capturing strategy, and it implies that the result is sensitive to whether a few large moves are captured.

Table 1: Headline performance, 1 September 2018 to 30 September 2026

	Portfolio 50/50	Bitcoin	Ether	Buy-and-hold
Final value of 100	6,536	1,481	11,590	1,073
CAGR	67.7%	39.6%	80.0%	34.1%
Annualised volatility	27.4%	25.0%	33.3%	69.0%
Sharpe ratio	2.02	1.46	1.93	0.78
Sortino ratio	4.73	3.23	4.62	–
Maximum drawdown	−17.8%	−24.8%	−24.1%	−76.9%
Calmar ratio	3.79	1.60	3.31	0.44
Skewness of daily returns	3.65	4.25	4.09	–
Trades (long / short)	716 (466 / 250)	358 (222 / 136)	358 (244 / 114)	–
Trades per year	89	44	44	–
Win rate	31%	30%	33%	–
Average win / average loss	5.58	5.18	5.52	–
Profit factor	2.59	1.93	2.73	–
Median holding time (hours)	13.0	11.5	14.9	–
Time in market	25%	15%	17%	100%

Historical simulation after modelled transaction costs; not live performance. Daily UTC close-to-close returns, annualised with $\sqrt{365.25}$, zero risk-free rate. Commission 0.045% per order; no slippage, funding or leverage. Bitcoin and Ether columns are the two halves of the portfolio, each measured on its own capital. Buy-and-hold: half of the capital in each coin on 1 September 2018, no rebalancing, no costs. Win rate, payoff and profit factor are computed on closed trades; the portfolio column pools the trades of both markets.

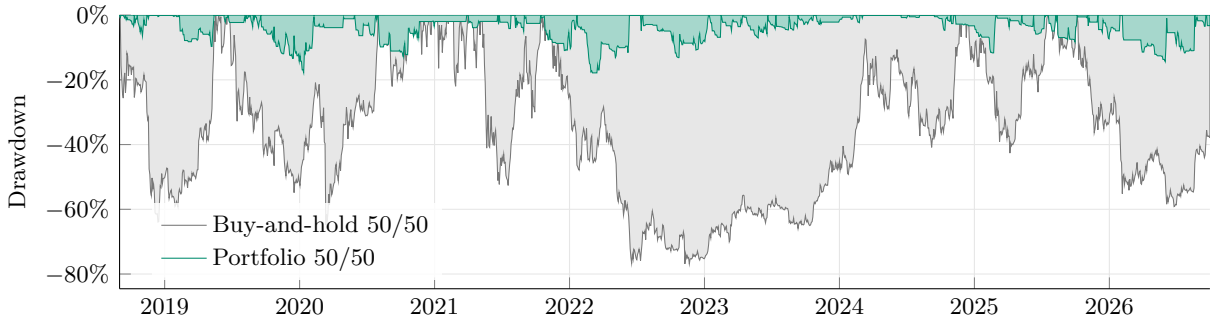


Figure 2: Drawdowns from the running peak, daily closes. Maximum: −17.8% for the portfolio, −76.9% for buy-and-hold.

6.2 Drawdowns and stability

The deepest portfolio drawdown is −17.8%, against −76.9% for buy-and-hold (Figure 2). Split in two halves, the portfolio Sharpe ratio is 2.12 (Sep. 2018–Sep. 2022) and 1.98 (Sep. 2022–Sep. 2026); the second half is weaker and the correlation between the two markets' strategy returns rises from 0.31 to 0.44. The rolling one-year Sharpe ratio (Figure 3) is positive in 100% of windows and above one in 93%, with a minimum of 0.27. A month-block bootstrap of 5,000 resampled histories gives a 90% interval of 1.53 to 2.51 for the Sharpe ratio; a bootstrap resamples history and is not a forecast. Calendar-year results are in Table 2.

6.3 Diversification

Bitcoin and Ether prices are highly correlated (daily correlation 0.82), but the two strategies much less (0.36): each is out of the market most of the time and their trades rarely coincide. The portfolio Sharpe ratio exceeds that of either market alone. In seven named market stress

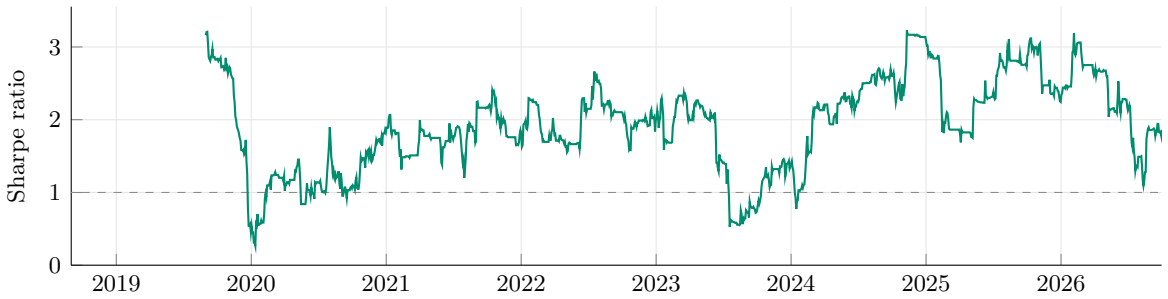


Figure 3: Rolling 365-day Sharpe ratio of the portfolio (2,588 windows). Positive in 100% of windows, above one (dashed) in 93%; median 1.94, minimum 0.27.

Table 2: Calendar-year returns

Year	Bitcoin	Ether	Portfolio	Sharpe	Max. drawdown	Buy-and-hold
2018[a]	16.2%	224.3%	120.3%	5.22	−6.4%	−50.3%
2019	55.5%	−9.0%	8.0%	0.45	−14.3%	49.0%
2020	21.2%	77.5%	56.1%	1.82	−12.8%	355.8%
2021	28.3%	60.4%	50.9%	1.80	−11.1%	191.2%
2022	40.1%	94.5%	80.9%	2.00	−17.8%	−66.4%
2023	59.3%	22.3%	29.5%	1.35	−7.1%	113.9%
2024	105.2%	66.3%	75.6%	3.13	−4.7%	78.1%
2025	1.2%	112.7%	81.6%	2.24	−8.6%	−8.5%
2026[a]	13.8%	63.9%	56.1%	1.95	−14.3%	−6.8%

[a] Partial year (1 September 2018 to 31 December 2018; 1 January 2026 to 30 September 2026). Sharpe ratio and maximum drawdown are those of the portfolio within the year. Buy-and-hold is the 50/50 benchmark of Table 1.

Table 3: Transaction costs, slippage and funding (portfolio 50/50)

Scenario	Sharpe	CAGR	Max. drawdown
Commission per order			
0.000%	2.17	74.5%	−17.2%
0.045% (base case)	2.02	67.7%	−17.8%
0.090%	1.88	61.2%	−19.7%
Slippage per order, on top of the base commission			
0.01%	1.99	–	–
0.02%	1.96	–	–
0.05%	1.86	–	–
Perpetual funding, on top of the base commission			
Historical perpetual funding (Binance)	1.97	64.7%	−17.9%
with Hyperliquid funding from May 2023	1.95	64.0%	−17.9%
Stress: shorts pay 0.01% per 8 hours throughout	2.00	66.7%	−18.1%

Funding is applied with its sign to long and short positions held across funding times; before 2020 a constant 0.01% per 8 hours is assumed. Market impact and capacity are not modelled here.

episodes defined from spot prices only, the portfolio was positive in 6 of 7.

6.4 Frictions

Turnover is high, about 87 times the capital a year, so costs matter. Doubling commissions, which is equivalent to adding 0.045% of slippage on every fill, lowers the Sharpe ratio to 1.88. Historical perpetual funding, with the correct sign for long and short positions, lowers it to 1.97 (Table 3). Capacity and market impact are not modelled here: the strategy enters precisely when liquidity is thinnest, which is the first thing a larger account would pay for. A separate study replays every execution since 2020 on real Binance perpetual-futures trades and reports the measured execution costs and capacity (institutional dossier).

7 Robustness

7.1 Placebos, delays and transfer across markets

For each market we fix pass/fail criteria before the result is seen: a positive event study of the signals and of shocks in the direction of the trend, a profit factor and a Sharpe ratio above preset levels, a rank above the 95th percentile of 200 random-entry runs that keep the exits identical, and a minimum share of the average trade retained when every entry is delayed by one bar. Table 4 reports the result.

The random-entry placebo is the most informative test: with the exits held fixed, random entry times do much worse, so the edge is not a by-product of market drift or of the exit structure. Delaying every entry by one bar keeps most of the average trade, so the result does not depend on a perfect fill. Ether, never used to calibrate the parameters (although the two entry conditions were set with knowledge of it), passes on two independent feeds. On the other markets they are partial on Solana and gold (strategy criteria met, event study not) and fail on TAO, and we report these results: the rules are not a universal law of markets, and we see no reason to hide it.

Table 4: Pre-set criteria by market, rules calibrated on Bitcoin only

Market	Feed	Period	Trades	Sharpe	CAGR	Max. DD	Random	Delay	Verdict
Bitcoin	Bitstamp	2017–2026	396	1.59	41.0%	−26.5%	100%	97%	pass (in-sample)
Ether	Binance	2018–2026	358	2.05	80.0%	−26.7%	100%	94%	pass
Ether	Dukascopy	2019–2026	206	1.82	57.5%	−23.6%	100%	73%	pass
Solana	Binance	2021–2026	251	0.57	16.0%	−54.0%	93%	80%	partial
Gold	Dukascopy	2013–2026	319	0.39	1.8%	−11.8%	100%	115%	partial
TAO	Binance	2024–2026	95	0.01	−7.3%	−39.2%	55%	n.m.	fail

Each market is simulated over its full history with the Bitcoin rules unchanged. Sharpe ratios here are annualised from 15-minute returns and differ slightly from the daily figures of Table 1. *Random*: share of 200 random-entry runs with identical exits that the strategy beats (criterion: above 95%). *Delay*: share of the average trade kept when every entry is delayed by one bar (n.m.: not meaningful, average trade not positive). *Verdict*: pass when the event study, profit factor, Sharpe, random-entry and delay criteria are all met; partial when the strategy criteria are met without the event study, or when only the Sharpe and random-entry criteria are met.

Table 5: Parameter neighbourhoods: all parameters perturbed jointly, 300 neighbours per level

Market	Perturbation	Median Sharpe	P10–P90	Chosen	Rank of chosen	Profitable
Bitcoin	±5%	1.49	1.42–1.58	1.59	92%	100%
	±10%	1.42	1.32–1.52	1.59	99%	100%
	±20%	1.35	1.23–1.47	1.59	100%	100%
Ether	±5%	2.05	1.91–2.19	2.05	55%	100%
	±10%	2.03	1.81–2.21	2.05	55%	100%
	±20%	1.94	1.76–2.14	2.05	76%	100%

Sharpe ratios annualised from 15-minute returns over each market's full history. *Rank of chosen*: percentile of the chosen parameters within their neighbourhood. *Profitable*: share of neighbours with a positive total return.

7.2 Parameter neighbourhoods

All parameters are perturbed together by up to ±5%, ±10% and ±20%, with 300 random neighbours per level (Table 5). Every neighbour is profitable on both markets. On Bitcoin the chosen parameters sit at the top of their neighbourhood, as expected of rules selected on that data; on Ether they sit in the middle, a plateau rather than a peak.

7.3 Walk-forward

On Bitcoin, re-selecting the shock threshold, stop and trailing-stop settings of the calm regime every quarter on the previous 36 months only, choosing plateaus rather than isolated optima, over 28 windows from 2020 to 2026, gives a Sharpe ratio of 0.83 (15-minute returns, 390 trades), against 1.55 for the fixed settings over the same windows. The strategy remains profitable out of sample, but clearly below the fixed preset: a large part of the headline performance on Bitcoin is a selection effect, and the forward test of Section 11 is the right place to measure it.

8 Selection risk of the rule choice

The calm-regime restriction was chosen among rule variants that were all replayed on the same portfolio. To measure how much of its advantage is selection, we take the 16 variants actually tried on the portfolio with the current parameters (short-entry rules, range filters, exit changes, early invalidation and agitated-regime alternatives) and apply combinatorially symmetric cross-

validation (Bailey et al., 2017): the history is cut into 16 blocks, and over all 12,870 ways of choosing 8 of them, the variant with the best Sharpe ratio on those blocks is ranked against all variants on the other 8. The chosen variant ranks above the median out of sample in 81% of splits, a probability of backtest overfitting of 19%. The current rules are the in-sample winner in 57% of splits and rank on average at the 89th percentile out of sample; they beat the earlier rules in 8 of 9 calendar years.

The size of the gain is another matter. The daily return difference between the current and the earlier rules has an annualised information ratio of 0.56; without any correction its one-sided probabilistic Sharpe ratio is 93.7%, and deflated for the 15 alternatives tried (Bailey and López de Prado, 2014) it falls to 18% (48% with doubled commissions). A chronological selection, re-made each January on past data only, would have chosen the current rules every year from 2020, for an out-of-sample Sharpe gain of 0.16 (95% interval -0.03 to 0.33). The choice is consistent over time, but its advantage over the earlier rules is not demonstrated beyond selection.

Deflating the absolute Sharpe ratio for the whole parameter search is more severe. With the dispersion of Sharpe ratios among the 206 random parameter sets from which the current ones were selected (standard deviation 0.60), the deflated Sharpe ratio falls below 95% beyond about 109 independent trials, and is 88% at 222. The search involved more trials than that, although the parameters were selected on Bitcoin only and the trials are far from independent. The historical Sharpe ratio should therefore not be read as an estimate of future performance.

9 Falsification of the short-entry condition

The conditional mirror is evaluated on the same history that was used to build the strategy, so its in-sample improvement is not evidence by itself. We pre-registered three tests asking whether the historical advantage is exceptional relative to counterfactuals with comparable properties. The condition itself is frozen; no parameter was changed after the results. The tests were run before the calm-regime restriction, on rules that also traded agitated-regime longs. Every short entry comes from the calm regime under both rule sets, so the short-trade statistics carry over; the portfolio-level statistics include those longs.

Test A: random selection. Keep exactly as many of the original short trades as the condition keeps, chosen at random within each market and year, 50,000 times. The average risk-normalised return of the shorts actually kept ranks at the 96.8th percentile of the draws ($p = 0.032$); the portfolio Sharpe ratio at the 94.5th percentile ($p = 0.055$).

Test B: shifted regimes. Shift the bearish periods circularly in time, keeping their persistence and their share of time, with a minimum shift of 180 days (2,593 shifts). The contrast Δ between shorts inside and outside the bearish regime is 0.436 with the real dates, larger than with 99.6% of the shifts ($p = 0.005$ for the portfolio, 0.093 for Bitcoin and 0.001 for Ether).

Test C: synthetic markets. Resample calendar blocks of 1, 3, 7, 14 and 30 days with replacement, the same blocks for both coins, rebuild synthetic 15-minute price paths, and recompute the engine, the regimes and the condition on each path (1,000 paths per block length). A generic effect of the condition would show up as a high probability of improvement on synthetic markets; an exceptional one as a historical improvement above the synthetic 95th percentile. Figure 4 shows the change in the average risk-normalised short return and Table 6 the details.

Reading. On synthetic markets the condition improves the average short in only about half of the paths, so its benefit to the quality of shorts is not mechanical; the historical improvement

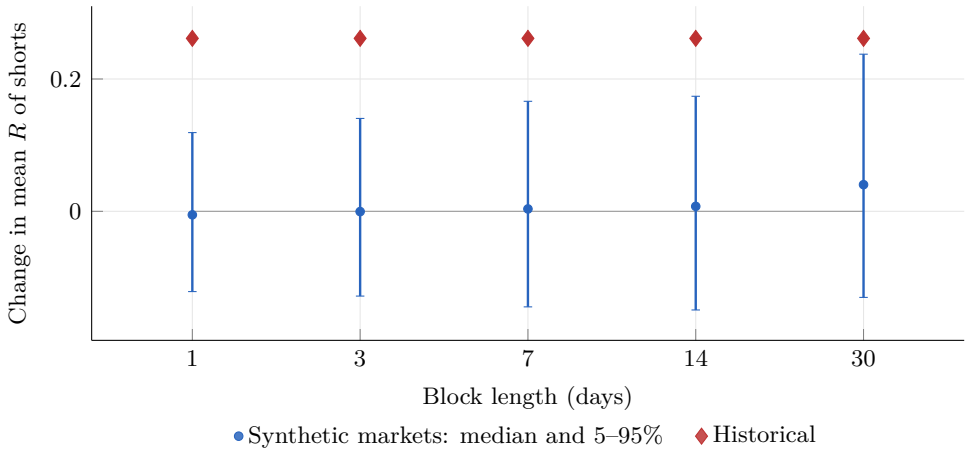


Figure 4: Test C. Change in the average risk-normalised return of short trades when the trend condition is applied, on 1,000 synthetic markets per block length and on the historical path. Historical percentile: 99.7% ($L = 1$), 99.6% ($L = 3$), 99.2% ($L = 7$), 99.4% ($L = 14$), 97.0% ($L = 30$).

Table 6: Test C by block length: synthetic markets against the historical path

Block (days)	$P(\Delta\text{EV} > 0)$	Synthetic P95	Historical	Hist. percentile	Sharpe gain, hist. pct.
1	0.47	0.119	0.261	99.7%	42.6%
3	0.50	0.140	0.261	99.6%	52.6%
7	0.51	0.166	0.261	99.2%	68.6%
14	0.53	0.174	0.261	99.4%	74.8%
30	0.66	0.238	0.261	97.0%	78.4%

ΔEV: change in the mean risk-normalised return of shorts when the condition is applied (portfolio). *Sharpe gain, hist. pct.*: percentile of the historical change in portfolio Sharpe ratio among synthetic paths. The condition improved the maximum drawdown in 97%, 94%, 89%, 87%, 83% of synthetic paths for $L = 1, 3, 7, 14, 30$ days.

Table 7: Residual alpha of the portfolio: regressions of daily returns on benchmark strategies

	M0	M1	M2 (pre-set)
α (annualised)	53.8%	44.2%	44.8%
	(4.85)	(4.60)	(4.94)
Buy-and-hold Bitcoin	−0.054	−0.043	−0.035
	(−1.89)	(−1.67)	(−1.41)
Buy-and-hold Ether	0.069	0.065	0.069
	(2.48)	(2.58)	(2.81)
TSMOM 30 days		0.095	0.073
		(3.11)	(2.52)
TSMOM 90 days		0.039	0.043
		(1.43)	(1.18)
TSMOM 180 days		−0.055	−0.070
		(−1.90)	(−2.44)
Breakout 55/20 days		0.073	0.048
		(1.84)	(1.22)
Moving averages 20/100 days		0.010	0.040
		(0.31)	(1.36)
Intraday breakout 24h/12h			0.150
			(5.59)
R^2	0.016	0.151	0.251
Residual information ratio	1.98	1.75	1.89
Share of mean return unexplained	97%	80%	81%
Bootstrap $P(\alpha \le 0)$	–	–	0.0000
Bootstrap 90% interval of α	–	–	29.7% to 60.0%

2,952 daily observations, 1 September 2018 to 30 September 2026. Newey–West t-statistics in parentheses (lag 8). M0: buy-and-hold only; M1: adds five daily trend strategies; M2: adds the intraday breakout. Bootstrap: 5,000 resamples of 97 calendar months, the same months for all series. Pre-set verdict for M2: $\alpha > 0$, $t \ge 3$ and $P(\alpha \le 0) < 0.01$.

exceeds the synthetic 95th percentile at every block length. By contrast, the gains in Sharpe ratio and drawdown are largely generic: removing shorts outside bearish regimes improves drawdowns in most synthetic paths as well. These tests use the same history that was used to build the strategy; they measure how unusual the result is, not how it will behave.

10 Is it just trend following? Residual alpha

A natural objection is that a strategy entering in the direction of large moves is a repackaged trend follower. We regress the portfolio's daily returns on buy-and-hold Bitcoin and Ether and on six standard trend strategies whose parameters were fixed before the study: time-series momentum over 30, 90 and 180 days, a 55/20-day breakout, a 20/100-day moving-average crossover and a 24h/12h breakout on 15-minute bars (Appendix A). The benchmarks use the same bars and commissions, positions decided at the close and applied to the next period. The pre-set verdict required $\hat{\alpha} > 0$, a Newey–West $t \ge 3$ (Harvey et al., 2016) and fewer than 1% of bootstrap histories with $\hat{\alpha} \le 0$. Table 7 reports three nested models, Table 8 the benchmarks on their own and Figure 5 the loadings of the pre-set model.

The residual alpha is 44.8% a year with $t = 4.94$; no bootstrap history out of 5,000 produced $\hat{\alpha} \le 0$, and the 90% interval is 29.7% to 60.0%. The factors explain $R^2 = 0.25$ of the daily variance and leave 81% of the average return unexplained. The largest loading is on intraday

Table 8: Benchmark strategies on their own, and sensitivity of the residual alpha

Benchmark (50/50, after costs)	Sharpe	CAGR	Max. drawdown	Corr. with SE
Buy-and-hold 50/50	0.82	37.7%	−76.3%	0.09
TSMOM 30 days	0.96	50.9%	−52.2%	0.34
TSMOM 90 days	0.45	7.9%	−90.6%	0.20
TSMOM 180 days	0.36	2.7%	−89.3%	0.08
Breakout 55/20 days	0.84	35.4%	−59.1%	0.33
Moving averages 20/100 days	0.64	22.6%	−69.9%	0.20
Intraday breakout 24h/12h	0.02	−14.8%	−88.6%	0.36
Shock Engine portfolio	2.02	67.7%	−17.8%	1.00
Sensitivity of M2	α	t	R^2	IR
19 trend strategies (full grid)	44.3%	5.07	0.325	1.97
First half (Sep. 2018–Sep. 2022)	48.6%	3.39	0.202	1.85
Second half (Sep. 2022–Sep. 2026)	44.1%	4.41	0.405	2.28
Weekly returns	45.7%	4.70	0.325	1.75
Benchmarks without costs	40.2%	4.62	0.251	1.70

Benchmarks average the two markets with daily rebalancing and pay the same commission as Shock Engine (0.045% per order); buy-and-hold carries no costs. Corr. with SE: correlation of daily returns with the Shock Engine portfolio. Definitions in Appendix A.

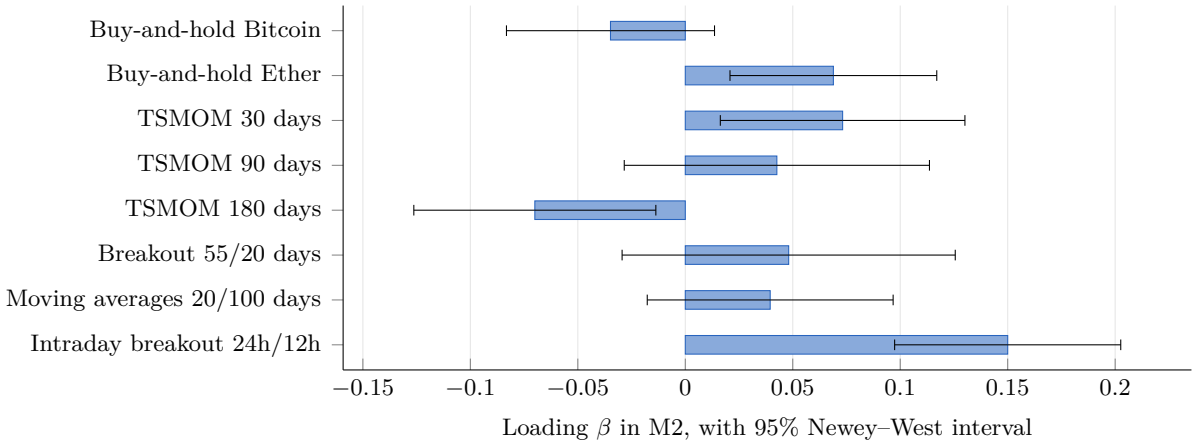


Figure 5: Factor loadings of the portfolio in the pre-set model M2. The largest loading is on intraday breakout 24h/12h ($t = 5.59$), which barely breaks even (Sharpe 0.02) on its own after costs.

breakout 24h/12h, which shares part of the timing of the strategy but barely breaks even (Sharpe 0.02) on its own after costs. Using 19 trend strategies at once, which deliberately favours the benchmarks, leaves the conclusion unchanged ($t = 5.07$). The verdict is "residual alpha demonstrated, in-sample": simple trend strategies do not reproduce the historical result. The regression cannot tell whether the alpha will persist.

11 Forward validation and live deployment

Pre-registered forward test. The hypothesis $H_1 : \mathbb{E}[R \mid T = 1] > \mathbb{E}[R \mid T = 0]$ for short signals is tested prospectively from 12 October 2026, with the definition, statistic, thresholds and stopping date committed before the first observation. Interim analyses occur when 25, 50 and 75 shorts have closed in the bearish regime, each at $p < 0.001$; the final analysis occurs when both groups reach 100 closed shorts, at $p < 0.047$, and no later than 1 October 2031, for a total type-I error of at most 0.05 (Haybittle, 1971; Peto et al., 1976). With roughly 89 trades a year across both markets, few of them shorts, the test will take years rather than months.

Why live Sharpe ratios take time. By Equation (5), a Sharpe ratio measured over one year of daily returns has a standard error of about 1.0: for a true Sharpe ratio of 1.5, the 95% interval of a one-year estimate runs from about -0.5 to 3.5, so a single year of live trading cannot reliably distinguish it from zero. What a first year can establish is *implementation fidelity*: that the live system takes the same decisions as the simulation on the same bars, at a measurable execution cost.

Deployment. The live engine shares its decision code with the backtester and is tested for decision-by-decision parity. Since 9 October 2026 a bot has run it on Hyperliquid's test network, where every order is publicly timestamped by the exchange; it applies the short-entry condition but not yet the calm-regime restriction; testnet funds are fictitious and its market differs from the real one, so these results check mechanics, not performance.

12 Limitations

- **Selection.** The parameters were selected on 2017–2026 Bitcoin data, which contains the whole portfolio period; Ether is a transfer across markets, not across time. The walk-forward Sharpe ratio (0.83) is clearly below the fixed preset, and the deflated Sharpe ratio does not survive the full parameter search.
- **Entry conditions in-sample.** The calm-regime restriction and the short-entry condition were chosen with knowledge of both markets; Sections 8 and 9 measure the selection risk and how unusual the short condition's benefit is, but neither makes them out-of-sample.
- **Frictions.** Slippage and funding are excluded from the headline figures. Capacity and market impact are not modelled here; execution costs are measured on Binance in a separate study, while the depth of Hyperliquid, the intended venue, is not yet measured. The strategy trades when order books are thinnest.
- **Right tail.** A small number of large trades drive the long-run growth; missing a few of them in live trading would change the outcome materially.
- **Venues.** Data venues (Bitstamp, Binance spot) differ from the intended execution venue (Hyperliquid perpetuals) in prices, fees, funding and stop mechanics.

- **Multiple testing.** Many tests are reported. Their criteria were fixed in advance and failures are reported, but the overall research process still involved choices that no single test accounts for.

13 Conclusion

Shock Engine trades the continuation of qualified 15-minute shocks in Bitcoin and Ether, entering only in calm volatility regimes, with a short side that mirrors the long side only in bearish daily regimes. On eight years of data the 50/50 portfolio delivers a Sharpe ratio of 2.02 with a maximum drawdown of -17.8%, survives placebo, delay, cost, parameter and cross-asset tests, and is not spanned by standard trend strategies. The same evidence is in-sample, depends on a convex right tail, and leaves capacity to a separate execution study. The decisive tests are the pre-registered forward analysis and a live track record, both under way and reported as they arrive, whatever they show.

A Benchmark strategies

For each market, a position $q_t \in \{-1, 0, 1\}$ is decided at the close of period t with information up to t and earns $q_t\,(P_{t+1}/P_t - 1) - c\,|q_t - q_{t-1}|$ with $c = 0.045\%$. Portfolio factors average the two markets with daily rebalancing; buy-and-hold Bitcoin and Ether enter separately.

- **Time-series momentum** TSMOM(L): $q_t = \text{sign}(P_t/P_{t-L} - 1)$, $L \in \{30, 90, 180\}$ days.
- **Breakout** (55/20 days): from flat, long if $P_t > \max_{t-55 \le s < t} P_s$ and short if $P_t < \min_{t-55 \le s < t} P_s$; a long exits when $P_t < \min_{t-20 \le s < t} P_s$ and a short when $P_t > \max_{t-20 \le s < t} P_s$.
- **Moving averages** (20/100 days): $q_t = +1$ if $\text{EMA}_{20}(t) > \text{EMA}_{100}(t)$, -1 otherwise.
- **Intraday breakout** (15-minute bars): the breakout rule with 96-bar entry and 48-bar exit windows (24 and 12 hours).

Long-only versions are not added: with positions ± 1 their returns are, up to costs, the average of the long/short version and buy-and-hold, already in the span of the regressors.

B Synthetic markets (Test C)

Calendar blocks of L days are drawn with replacement from the full history, the same blocks for both coins. Within a block, 15-minute bars are rescaled so that the synthetic path is continuous at block boundaries; volumes are normalised by their trailing mean; the price grid follows a fixed rule. The engine, the volatility regimes and the trend condition are recomputed from scratch on each path. On the identity path the construction reproduces the rules under test exactly (identical regimes and equity bar by bar), which validates the pipeline.

C Reproducibility

Every number in this paper is produced by versioned code from versioned data and read from the published research files (research version `shock-engine-2026.10.13`, code commit `f056418`, parameter hash `c6ac30464bce`); the portfolio pipeline runs 39 consistency checks, all passing. Pre-registration documents were committed before the corresponding computations. Data files, daily returns and the full numerical report are available at https://www.shadowmarketpro.com/research.

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